

SQL PROJECT



UNLOCKING SALES GROWTH: PIZZA HOUSE DATA ANALYSIS

AN END-TO-END SQL PORTFOLIO PROJECT

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Project Objective



The Goal :

To analyze a year's worth of sales data to identify peak operational hours, revenue drivers, and customer preferences.

The Impact :

Provide actionable, data-driven recommendations to optimize inventory, marketing spend, and staff scheduling.





Tech Stack

Tools Used :

- MySQL
- Canva

Dataset :

Provide actionable, data-driven recommendations to optimize inventory, marketing spend, and staff scheduling.

Key Techniques:

Complex Joins, Window Functions (CTEs), Subqueries, Aggregate Functions.



Data Preparation & Import

01.

Received raw pizza sales data spread across four flat .csv files (Orders, Order Details, Pizzas, Pizza Types).

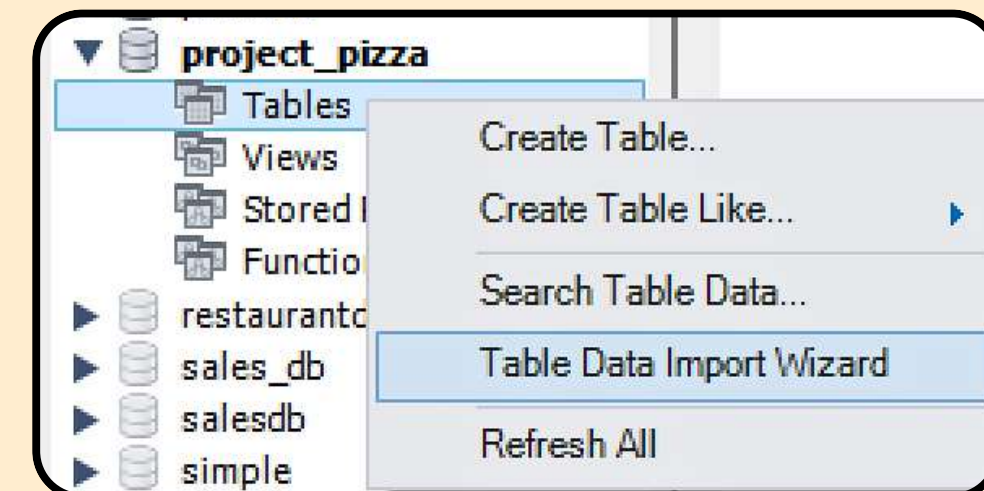
02.

Authored CREATE TABLE scripts in MySQL to build a structured relational schema, strictly defining data types, primary keys, and foreign keys for data integrity.

03.

Successfully imported over 21,000+ rows of flat data into the SQL database, paving the way for complex, multi-table queries.

Name	Status
 order_details.csv	
 orders.csv	
 pizza_types.csv	
 pizzas.csv	



Result Grid			
	order_id	order_date	order_time
▶	1	2015-01-01	11:38:36
	2	2015-01-01	11:57:40
	3	2015-01-01	12:12:28
	4	2015-01-01	12:16:31
	5	2015-01-01	12:21:30
	6	2015-01-01	12:29:36
	7	2015-01-01	12:50:37
	8	2015-01-01	12:51:37
	9	2015-01-01	12:52:01
	10	2015-01-01	13:00:15
	11	2015-01-01	13:02:59
	12	2015-01-01	13:04:41

orders 8 ×

Result Grid				
	pizza_type_id	name	category	ingredients
▶	bbq_ckn	The Barbecue Chicken Pizza	Chicken	Barbecued Chicken, Red Peppers, Green Peppe...
	cali_ckn	The California Chicken Pizza	Chicken	Chicken, Artichoke, Spinach, Garlic, Jalapeno P...
	ckn_alfredo	The Chicken Alfredo Pizza	Chicken	Chicken, Red Onions, Red Peppers, Mushrooms...
	ckn_pesto	The Chicken Pesto Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Spinach, Garl...
	southw_ckn	The Southwest Chicken Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Red Onions, ...
	thai_ckn	The Thai Chicken Pizza	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, T...
	big_meat	The Big Meat Pizza	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sau...
	classic_dlx	The Classic Deluxe Pizza	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppe...
	hawaiian	The Hawaiian Pizza	Classic	Sliced Ham, Pineapple, Mozzarella Cheese
	ital_cpdlo	The Italian Capocollo Pizza	Classic	Capocollo, Red Peppers, Tomatoes, Goat Chee...
	napolitana	The Napolitana Pizza	Classic	Tomatoes, Anchovies, Green Olives, Red Onion...
	pep_msh_pep	The Pepperoni, Mushroom, ...	Classic	Pepperoni, Mushrooms, Green Peppers

pizza_types 7 ×

Result Grid				
	order_details_id	order_id	pizza_id	quantity
▶	1	1	hawaiian_m	1
	2	2	classic_dlx_m	1
	3	2	five_cheese_l	1
	4	2	ital_supr_l	1
	5	2	mexicana_m	1
	6	2	thai_ckn_l	1
	7	3	ital_supr_m	1
	8	3	prsc_argla_l	1
	9	4	ital_supr_m	1
	10	5	ital_supr_m	1
	11	6	bbq_ckn_s	1
	12	6	the_greek_s	1

order_details 9 ×

Result Grid				
	pizza_id	pizza_type_id	size	price
▶	bbq_ckn_s	bbq_ckn	S	12.75
	bbq_ckn_m	bbq_ckn	M	16.75
	bbq_ckn_l	bbq_ckn	L	20.75
	cali_ckn_s	cali_ckn	S	12.75
	cali_ckn_m	cali_ckn	M	16.75
	cali_ckn_l	cali_ckn	L	20.75
	ckn_alfredo_s	ckn_alfredo	S	12.75
	ckn_alfredo_m	ckn_alfredo	M	16.75
	ckn_alfredo_l	ckn_alfredo	L	20.75
	ckn_pesto_s	ckn_pesto	S	12.75
	ckn_pesto_m	ckn_pesto	M	16.75
	ckn_pesto_l	ckn_pesto	L	20.75

pizzas 6 ×

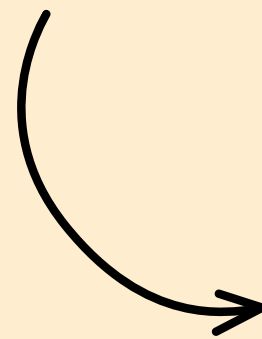


Our 4 Tables

Key Metrics

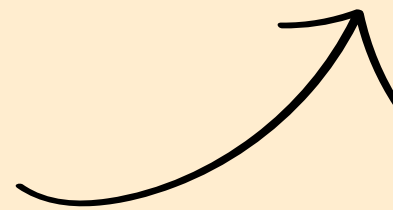
01.

Total Revenue:
\$818K+



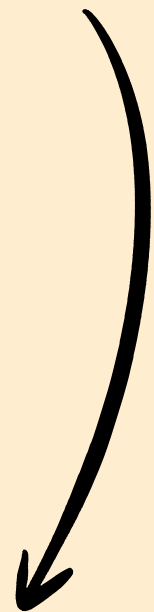
02.

Total Orders:
21,350



03.

Bestselling Category: Classic
Pizzas (26.9% of Revenue)



04.

Busiest Hours: 12:00 PM –
1:00 PM & 5:00 PM – 7:00 PM



Queries

- Basic •

02.

Retrieve the total number of orders placed.

```
-- Retrieve the total number of orders placed.  
SELECT COUNT(order_id) AS Total_Orders FROM orders;
```

Result Grid	
	Total_Orders
▶	21350

04.

Identify the highest-priced pizza.

```
-- Identify the highest-priced pizza.  
SELECT pt.name, p.size, p.price FROM pizza_types pt      -- (Method 1)  
JOIN pizzas p  
ON pt.pizza_type_id = p.pizza_type_id  
ORDER BY p.price DESC  
LIMIT 1;  
  
SELECT pt.name, p.size, p.price,                          -- (Method 2 - Just to check ranks)  
       RANK() OVER(ORDER BY p.price DESC) AS Highest_Price  
FROM pizza_types pt  
JOIN pizzas p  
ON pt.pizza_type_id = p.pizza_type_id;
```

	name	size	price	Highest
▶	The Greek Pizza	XXL	35.95	1
	The Greek Pizza	XL	25.5	2
	The Brie Carre Pizza	S	23.65	3
	The Italian Vegetables Pizza	L	21	4
	The Spinach Supreme Pizza	L	20.75	5
	The Barbecue Chicken Pizza	L	20.75	5
	The California Chicken Pizza	L	20.75	5
	The Spicy Italian Pizza	L	20.75	5

Result Grid			
	name	size	price
▶	The Greek Pizza	XXL	35.95

01.

Calculate the total revenue generated from pizza sales.

```
-- Calculate the total revenue generated from pizza sales.  
SELECT ROUND(SUM(p.price * o.quantity),2) AS Total_Amount  
FROM pizzas p  
JOIN order_details o  
ON p.pizza_id = o.pizza_id;
```

Result Grid	
	Total_Amount
▶	817860.05

03.

Identify the most common pizza size ordered.

```
-- Identify the most common pizza size ordered.  
SELECT p.size, COUNT(od.quantity) AS Quantity_Ordered  
FROM order_details od  
JOIN pizzas p  
ON od.pizza_id = p.pizza_id  
GROUP BY p.size  
ORDER BY Quantity_Ordered DESC  
LIMIT 1;
```

Result Grid		
	size	Quantity_Ordered
▶	L	18526

05.

List the top 5 most ordered pizza types along with their quantities.

```
-- List the top 5 most ordered pizza types along with their quantities.  
SELECT pt.name, SUM(od.quantity) AS Total_Quantity_Ordered      -- (Joined 3 Tables here)  
FROM pizza_types pt  
JOIN pizzas p  
ON pt.pizza_type_id = p.pizza_type_id  
JOIN order_details od  
ON od.pizza_id = p.pizza_id  
GROUP BY pt.name  
ORDER BY Total_Quantity_Ordered DESC  
LIMIT 5;
```

Result Grid

Filter Rows:

	name	Total_Quantity_Ordered
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

Queries

• Intermediate •



01.

Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT pt.category, SUM(od.quantity) AS Total_Quantity
FROM pizza_types pt
JOIN pizzas p
  ON pt.pizza_type_id = p.pizza_type_id
JOIN order_details od
  ON od.pizza_id = p.pizza_id
GROUP BY pt.category;
```

category	Total_Quantity
Classic	14888
Veggie	11649
Supreme	11987
Chicken	11050

02.

Determine the distribution of orders by hour of the day.

```
SELECT HOUR(order_time) AS Hour_of_the_Day, COUNT(order_id) FROM orders
GROUP BY Hour_of_the_Day
ORDER BY Hour_of_the_Day ASC;
```

Hour_of_the_Day	COUNT(order_id)
9	1
10	8
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336

03.

Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT category AS Category_Name, COUNT(name) AS Types_we_offer FROM pizza_types
GROUP BY category;
```

Category_Name	Types_we_offer
Chicken	6
Classic	8
Supreme	9
Veggie	9

04.

Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT o.order_date, SUM(od.quantity) AS Total_Orders,
       ROUND(AVG(SUM(od.quantity)) OVER() ,2)AS Average_Overall_For_Total_Orders
FROM orders o
JOIN order_details od
ON o.order_id = od.order_id
GROUP BY o.order_date;
```

order_date	Total_Orders	Average_Overall_For_Total_Orders
2015-01-01	162	138.47
2015-01-02	165	138.47
2015-01-03	158	138.47
2015-01-04	106	138.47
2015-01-05	125	138.47
2015-01-06	147	138.47
2015-01-07	138	138.47
2015-01-08	173	138.47
2015-01-09	127	138.47

05.

Determine the top 3 most ordered pizza types based on revenue.

```
SELECT pt.name, SUM(p.price * od.quantity) AS Revenue
FROM pizza_types pt
JOIN pizzas p
ON pt.pizza_type_id = p.pizza_type_id
JOIN order_details od
ON od.pizza_id = p.pizza_id
GROUP BY pt.name
ORDER BY Revenue DESC
LIMIT 3;
```

name	Revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Queries

• Advance •



01.

Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT pt.name,  
       ROUND(SUM(p.price * od.quantity) * 100.0 / SUM(SUM(p.price * od.quantity)) OVER(), 2) AS Percentage  
FROM pizza_types pt  
JOIN pizzas p ON pt.pizza_type_id = p.pizza_type_id  
JOIN order_details od ON od.pizza_id = p.pizza_id  
GROUP BY pt.name;
```

name	Percentage
The Hawaiian Pizza	3.95
The Classic Deluxe Pizza	4.67
The Five Cheese Pizza	3.19
The Italian Supreme Pizza	4.09
The Mexicana Pizza	3.27
The Thai Chicken Pizza	5.31
The Prosciutto and Arugula Pizza	2.96
The Barbecue Chicken Pizza	5.23
The Greek Pizza	3.48

02.

Analyze the cumulative revenue generated over time.

```
SELECT o.order_date,  
       ROUND(SUM(p.price * od.quantity),2) AS Revenue,  
       ROUND(SUM(SUM(p.price * od.quantity)) OVER(ORDER BY o.order_date),2) AS Running_Revenue_Over_Dates  
FROM orders o  
JOIN order_details od  
  ON o.order_id = od.order_id  
JOIN pizzas p  
  ON p.pizza_id = od.pizza_id  
GROUP BY o.order_date  
ORDER BY order date;
```

order_date	Revenue	Running_Revenue_Over_Dates
2015-01-01	2713.85	2713.85
2015-01-02	2731.9	5445.75
2015-01-03	2662.4	8108.15
2015-01-04	1755.45	9863.6
2015-01-05	2065.95	11929.55
2015-01-06	2428.95	14358.5
2015-01-07	2202.2	16560.7
2015-01-08	2838.35	19399.05
2015-01-09	2127.35	21526.4

03.

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
SELECT category, name, Total_Quantity, Revenue
FROM(
  SELECT category, name, Total_Quantity, Revenue,
    RANK() OVER(PARTITION BY category
      ORDER BY Revenue DESC)
    AS Rank_in_Category
  FROM(
    SELECT pt.category, pt.name, SUM(od.quantity) AS Total_Quantity, SUM(p.price * od.quantity) AS Revenue
    FROM pizza_types pt
    JOIN pizzas p
      ON pt.pizza_type_id = p.pizza_type_id
    JOIN order_details od
      ON od.pizza_id = p.pizza_id
    GROUP BY pt.category, pt.name
  ) AS x
) AS z
WHERE Rank_in_Category <= 3
ORDER BY category,Revenue DESC;
```

category	name	Total_Quantity	Revenue
Chicken	The Thai Chicken Pizza	2371	43434.25
Chicken	The Barbecue Chicken Pizza	2432	42768
Chicken	The California Chicken Pizza	2370	41409.5
Classic	The Classic Deluxe Pizza	2453	38180.5
Classic	The Hawaiian Pizza	2422	32273.25

Insights



01.

When Do Customers Order the Most?

➤ By extracting the hour from order timestamps and aggregating the volume, we identified massive traffic spikes during the lunch rush (12-1 PM) and dinner time (5-7 PM).

02.

Which Items Drive the Bottom Line?

➤ Using nested subqueries and complex joins, we calculated the percentage contribution of each pizza category. The Thai Chicken Pizza generates the highest overall revenue, despite Classic being the most ordered category.

03.

Sizing Up Customer Preferences

➤ Group-by aggregations revealed a strong preference for Large pizzas, which account for the vast majority of total sales volume.

04.

Tracking Cumulative Revenue

➤ To understand sales momentum, I utilized Window Functions to calculate the rolling cumulative revenue day-over-day.

Strategic Recommendations



01.

Staffing Optimization :

➤ Increase staff shifts exclusively between 12-1 PM and 5-7 PM to reduce customer wait times.

02.

Inventory Focus :

➤ Prioritize stocking ingredients for the Thai Chicken Pizza and Large-sized dough to prevent stockouts of high-margin items.

03.

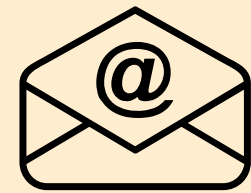
Marketing Opportunity :

➤ Run targeted promotions during the mid-afternoon slump (2 PM - 4 PM) to drive steady revenue throughout the day.

Thank You !

for reviewing my analysis

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[https://github.com/kumbharom1519/
SQL_Project_Pizza_Sales_Analysis](https://github.com/kumbharom1519/SQL_Project_Pizza_Sales_Analysis)

